

STAT117 Fall 2026 Exam #1

- Ask if you have any questions about the exam.
- Be concise in your answers. You will be penalized for adding irrelevant or incorrect information to a response.
- You don't need to round decimals, but if you choose to, keep at least two decimals unless otherwise noted.
- You are allowed a reference sheet. Use both sides of one 8.5 by 11 inch piece of paper and put whatever you want on it. You will turn this sheet in with your exam. Be sure to put your name on it so that it can be returned to you later.
- Calculator use is permitted. You may NOT use your phone as a calculator. You will need to use your laptop, but the only programs you are allowed to have open are Jamovi and the standard Windows or Mac calculator. You need to close (not just minimize) other programs (including your web browser and all messaging apps). You are not allowed to consult any papers (other than your reference sheet), websites, books, or any person except the instructor during the exam.
- You need to download the dataset before the exam. You are not allowed to open a web browser during the exam. You must turn off WIFI on your computer.
- Cases of suspected cheating will be referred to the dean. We take cheating very seriously because it benefits student learning when students get grades that reflect what they know. For this reason, you are allowed to have at your seat only your test, your reference sheet, your computer, a calculator, and writing implements (pen or pencils). You will need to put your jacket, backpack, cell phone, smart watch and anything else you carry to class at the front of the room during the exam.
- I might tell you where to sit, or move your seat, during the exam.
- The dataset used for the exam is **child_care.sav**.
 - Source: *Survey on Where Parents Look for and Find Information and How They Use Information When Selecting Child Care, United States, 2024* ([ICPSR 39220](#)).
 - **The levels of measurement in Jamovi are not set correctly, so you need to judge for yourself.** Sometimes, the level of measurement is subjective or controversial, and the relevant question will tell you what to do.
 - Values labelled “No response” (-9) and “Logical skip” (-8) are treated as missing values. You can ignore them and do not need to include them in calculations.

1. (20 points) We first examine some basic information about the dataset.

(a) How many variables are recorded in the dataset? How many respondents?

30 variables; 2035 respondents.

(b) Give an example of a dichotomous variable. Mention both the variable name and its description.

`respondent_gender` is dichotomous (Male/Female).

(c) Among respondents, what percentage identify as female (`respondent_gender`)? Do not round your answer; report at least one decimal place.

$1306/2035 = 64.1769\%$; 64.2%.

(d) What does the variable `respondent_education` represent? What is its mode? Provide an interpretation of the mode.

Respondent's highest degree earned.

Mode is 3, "Some college/associates degree."

Interpretation: this is the most frequently reported education category. It was selected by 674 of 2035 respondents (33.1%), not a majority.

2. (20 points) Answer the following questions about `care_decision_hours`, the number of hours spent making the most recent child care decision.

(a) Find the respondent with `case_id = 1`. How many hours did this respondent spend making the most recent child care decision? Report the corresponding percentile rank and interpret it in context.

10 hours; percentile rank = 71.3%. About 71.3% of respondents with a valid answer spent 10 hours or less making their most recent child care decision.

(b) Among respondents who answered this question, what percentage spent 5 hours or less making their most recent child care decision?

50.7%.

(c) Calculate the skewness ratio of `care_decision_hours`. Based on this value, describe the shape of the distribution.

$6.24/0.066 = 94.4$; the distribution is severely right-skewed.

(d) A boxplot of `care_decision_hours` shows points above the upper whisker. Why? Choose one.

- (i) There are too many missing values for this survey question.
- (ii) The maximum and third quartile are the same value.
- (iii) These points are considered outliers.
- (iv) Boxplots should not be used for numeric variables.

(iii)

3. (15 points) In this problem, compare `weekly_work_school_commute_hrs` (respondent work, school, and commuting hours in a typical week) between men and women (`respondent_gender`).

(a) Calculate the skewness ratios of weekly hours for men and women. Is either group's distribution severely skewed? If so, state the direction.

Men: $0.104/0.094 = 1.11$; skew is not severe. Women: $0.406/0.070 = 5.78$; severely right-skewed.

(b) Which group typically reports more work, school, and commuting hours per week? Support your answer with an appropriate statistic that is robust to skewness and outliers.

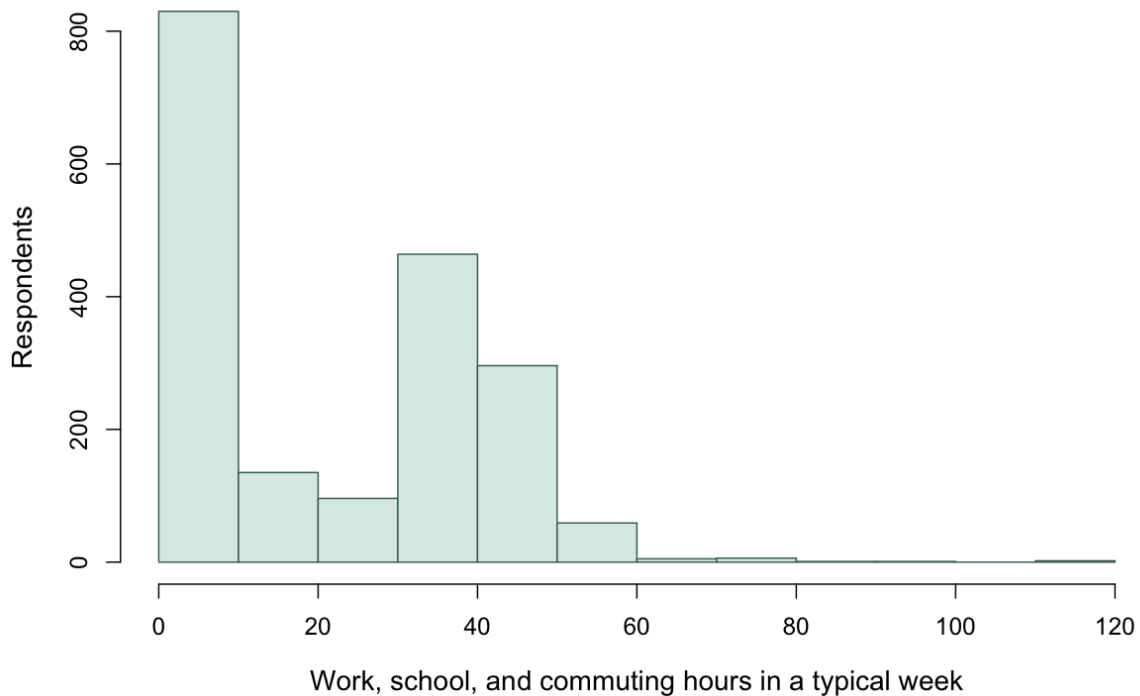
Men: median = 40 hours. Women: median = 16 hours. Men typically report more hours, based on the medians.

(c) Which group is less consistent in weekly hours? Support your answer with the name and value of an appropriate statistic that is robust to skewness and outliers.

Men: IQR = 42 hours. Women: IQR = 40 hours. Men are slightly less consistent because their IQR is larger.

4. (15 points) Answer the following questions about `weekly_work_school_commute_hrs` (respondent work, school, and commuting hours in a typical week).

(a) Given below is a histogram of `weekly_work_school_commute_hrs`.



Comment on the modality (unimodal, bimodal, multimodal) and skewness (symmetric, left-skewed, right-skewed) of the distribution. Provide a brief description *in the context* of respondents' weekly work, school, and commuting hours.

Bimodal and right-skewed: there is a cluster near 0-10 hours and another around 30-50 hours, with a long tail toward higher weekly hours.

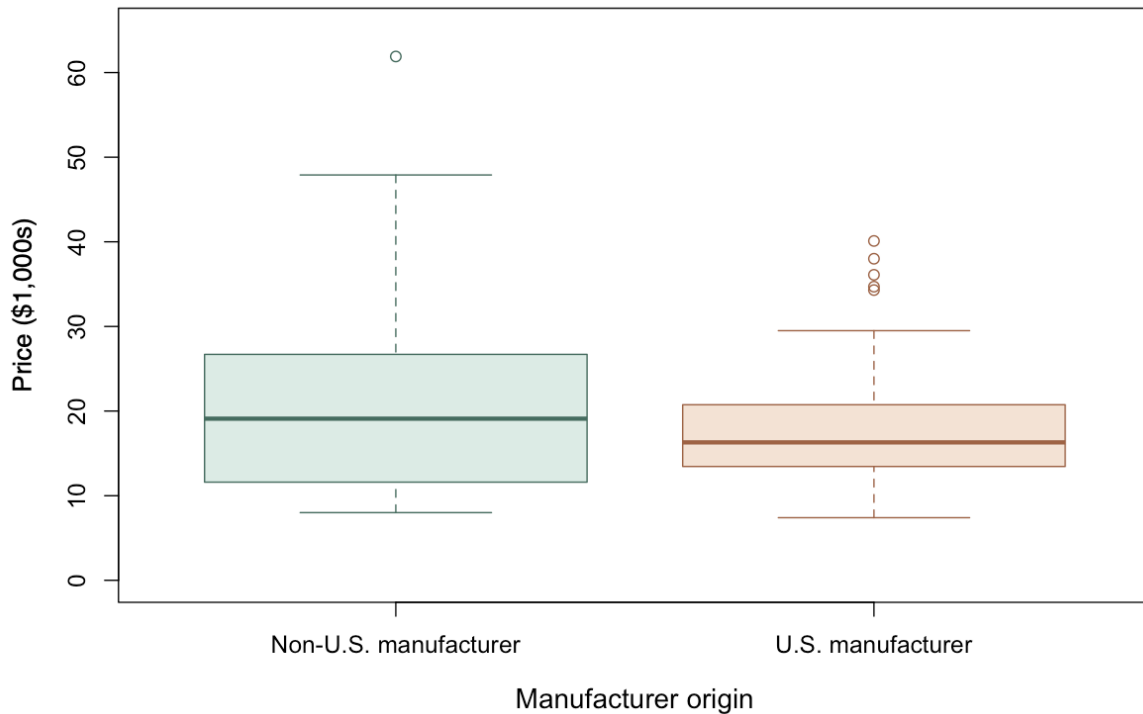
(b) Report the mean and standard deviation of `weekly_work_school_commute_hrs`.

Mean = 23.1 hours; SD = 20.5 hours.

(c) Define a new variable, `daily_hrs`, as the number of hours spent on work or school per day, by `daily_hrs = weekly_work_school_commute_hrs / 5 - 1` (dividing 5 because there are typically 5 work days in a week; subtracting 1 to account for an average 1-hour commute included in the original data). Calculate the mean and SD values for `daily_hrs`. Show your work for full credit.

$\text{daily_hrs} = \text{weekly_work_school_commute_hrs} / 5 - 1$; Mean = $23.1 / 5 - 1 = 3.62$ hours; SD = $20.5 / 5 = 4.1$ hours.

5. (20 points) You do not need Jamovi for this question. The boxplots below show the price (in thousands of dollars) for cars based on a [car dataset](#) that you do not have access to, grouped by whether the manufacturer is based in the United States.



(a) Estimate the median price for cars from non-U.S. manufacturers from the graph. Indicate on the graph how you arrived at your estimate.

About \$19,100.

(b) For U.S. manufacturers, does the distribution of prices appear symmetric, left-skewed, or right-skewed? Briefly explain.

Right-skewed; the upper whisker extends farther than the lower whisker, and several high-price outliers appear above it.

(c) Which manufacturer's cars typically have a higher price, U.S. or non-U.S.? Which part of the boxplots supports your answer? Name a relevant statistic.

Non-U.S. manufacturers, based on the median line: about \$19,100 compared with about \$16,300 for U.S. manufacturers.

(d) Which manufacturer group is more homogeneous in price? Which part of the boxplots did you use? Name a relevant statistic.

U.S. manufacturers, based on the IQR (the width of the box): about \$7,250 for U.S. manufacturers compared with about \$15,100 for non-U.S. manufacturers.

6. (10 points) You do not need Jamovi for this question. Use the information in a recent New York Times article [DoorDash Agrees to \\$131.5 Million Settlement for Shortchanging Workers](#). The article says:

“DoorDash has agreed to pay \$131.5 million in a settlement with New York City that allows closer monitoring of its payment methods after it underpaid or failed to pay 264,000 delivery workers.”

“DoorDash will pay \$115 million in worker relief in part for not adhering to a 2023 law that was enacted to improve the compensation of tens of thousands of delivery workers in New York, most of whom ride e-bikes and mopeds. The company will also pay about \$16 million in civil penalties.”

“DoorDash said the median payment to be returned to workers amounts to around \$48. City officials said about 700 workers are owed more than \$10,000 each; more than 27,000 workers are owed more than \$1,000; and more than 4,000 workers are owed more than \$5,000.”

(a) Which of the following is a correct interpretation of the reported median payment of \$48? Choose one.

- (i) The vast majority of workers (more than 90%) will receive \$48.
- (ii) Half of the workers will receive \$48 or less and half of the workers will receive \$48 or more.
- (iii) Every affected worker will receive exactly \$48.
- (iv) Every affected worker will receive at least \$48.

(ii)

(b) If all \$115 million in worker relief were divided equally among the 264,000 affected workers, about how much would each worker receive? How does this hypothetical average compare with the reported median payment of \$48? What does this tell you about the shape of the distribution?

$\$115,000,000 / 264,000 = \435.61 per worker. This hypothetical average is much higher than the reported median of about \$48, suggesting a strongly right-skewed distribution of payments, with much larger payments to some workers.